MAXIMAL SIGNED VOLUME FOR (MULTIVARIATE) SUPERMODULAR QUASI-COPULAS

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Abstract. Copulas are the main tool of dependence modeling in statistics and quasi-copulas are their necessary companions. The latter appear, say, as infima or suprema of sets of copulas, they form a huge class and have some unpleasant properties. Their statistical interpretation is challenged by the fact that they may lead to negative volumes of some boxes. So, numerous applications call for an intermediate class and supermodular quasi-copulas is one of them having many useful properties. An excellent measure, Average Rectangular Volume (ARV in short), to clarify and position this class was proposed in the seminal paper by Anzilli and Durante, The average rectangular volume induced by supermodular aggregation functions, *J. Math. Anal. Appl.* **555** (2026) 21 pp. While supermodularity is a bivariate notion, its extension to d-variate case for $d > 2$ was recently provided in a key paper by Arias-García, Mesiar, and De Baets, The unwalked path between quasi-copulas and copulas: Stepping stones in higher dimensions, *Int. J. of Appr. Reasoning*, **80** (2017) pp. 89–99. Here an alternative method to ARV is presented extendable to the multivariate case based on Maximal (in absolute value) Negative Volumes (MNV in short) on boxes thus helping practitioners when seeking the right (quasi-)copula for their problem. Observe that these volumes on copulas are zero, while their values on quasi-copulas, depending on d, has been a long-open problem solved only recently. We present a nontrivial extension of this solution to exhibit the main goal of this paper, a measure that clarifies and positions the classes considered based on MNV.

1. Introduction

Copula [1] is a joint distribution with uniform margins, but when Sklar [2] discovered in 1959 that we could get any distribution on earth when we inserted appropriate univariate distributions as margins into a copula, it became a universal device to solve dependence problems. There are many applications, though, where quasi-copulas are also needed, they form a pretty big class and have some unpleasant properties. In particular, we may get negative volumes for some boxes. The so motivated search for intermediate classes inspired by applications brings us to supermodular quasi-copulas that have many useful properties.

Supermodularity is a bivariate notion [4], see also [12, Definition 2.1], which coincides with the 2-increasing property. In dimensions $d \geq 2$, beyond the class of 2-increasing functions, a hierarchy of intermediate classes satisfying the k-increasing property for $k = 2, \ldots, d$ was uncovered by Arias-García et al. [5], extending the volume-based characterization of quasi-copulas to k-dimensional slices of the d-dimensional box. In this way one obtains special subclasses of quasi-copulas: they are strictly contained within the class of all quasi-copulas (corresponding to the 1-increasing property) and contain all copulas (corresponding to the d-increasing property). Consequently, the notion of k-increasingness can be viewed as a natural generalization of supermodularity to higher dimensions as we mentioned earlier. The main results of this paper are two types of measures that clarify and position quasi-copulas in each of these classes: the Maximal (in absolute value) Negative Volumes (MNV in short) on boxes and their counterpart Maximal Positive Volumes (MPV in short) what brings us to their tie Maximal Signed Volumes (MSV for short). This technique extends an excellent bivariate method, Average Rectangular Volume (ARV in short), that aggregates the knowledge contained in a supermodular quasi-copula and was proposed by Anzilli and Durante in their seminal paper [3] in 2026.

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As there seems to be no obvious extension of the method presented in [3] to higher dimensions, we develop a novel method based on maximal (in absolute value) negative volume on boxes. While this value on copulas is zero, its value on quasi-copulas depends on d non-trivially. Here is brief story of its discovery. The seminal paper [6] by Arias-García et al. which is in some sense a 2020 sequel of [5] presents a survey of the most significant results and open problems in quasi-copula theory. Their Open Problem #5 asks for the maximal negative and maximal positive mass *over all boxes and over all quasi-copulas*, for the characterization of the boxes attaining these extremes, and for characterization of quasi-copulas with this property. Here, *maximal negative* refers to the negative volume with the largest absolute magnitude. The problem seems to have been first proposed in 2002 by Nelsen et al. [7], the trivariate solution was first presented in 2007 by De Beats et al. [8] and the problem was solved in papers [9] in 2023 and [10, 11] in 2025/26.

The manuscript is organized as follows. In Section 2 we present our main results (Theorems 1 and 2), together with a numerical analysis of the extremal values (Tables 1 and 2). Section 3 formulates the maximal-volume problems as linear programs (Proposition 1), simplifies them using a symmetrization argument (Corollary 1), introduces new sets of variables (Proposition 3), and derives the dual linear programs (Proposition 4). In Section 4 we prove Theorems 1 and 2 by solving these dual programs exactly. Finally, in Section 5 we summarize the paper and give some directions for further work.

2. Statements of the main theorems

In this section we introduce the notation and state our main results, i.e., Theorem 1 solves the maximal negative volume problem for k-increasing quasi-copulas, while Theorem 2 solves the maximal positive volume problem. Concrete numerical solution examples are also presented in the form of tables. (See Tables 1 and 2 for the minimal and the maximal volume question, respectively.)

Let $\mathcal{D} \subseteq [0,1]^d$ be a non-empty set and $d \in \mathbb{N}$, $d \geq 2$. We say that a function $F : \mathcal{D} \to [0,1]$:

- is *d-increasing* if for any d-box $\mathcal{B} := \prod_{i=1}^d [a_i, b_i] \subseteq \mathcal{D}$ it holds that
$$V_F(\mathcal{B}) := \sum_{z \in \mathrm{Vert}(\mathcal{B})} (-1)^{S(z)}\, F(z) \ \geq 0,$$
where $\mathrm{Vert}(\mathcal{B})$ stands for the vertices of $\mathcal{B}$ and
$$S(z) = \mathrm{card}\{\, j \in \{1, 2, \ldots, d\} \colon z_j = a_j \,\}.$$
The quantity $V_F(\mathcal{B})$ is called the *F-volume* of $\mathcal{B}$. Here $\mathrm{card}\, A$ denotes the cardinality of the set A.
- satisfies the *boundary condition* if for $\underline{u} := (u_1, \ldots, u_d) \in \mathcal{D}$ the following hold:
$$\begin{aligned} &(a) \quad \text{If } \underline{u} = (u_1, \ldots, u_{i-1}, 0, u_{i+1}, \ldots, u_d) \text{ for some } i, \text{then } F(\underline{u}) = 0. \\ &(b) \quad \text{If } \underline{u} = (1, \ldots, 1, u_i, 1, \ldots, 1) \text{ for some } i, \text{then } F(\underline{u}) = u_i. \end{aligned}$$
- satisfies the *monotonicity condition* if it is nondecreasing in every variable, i.e., for each $i = 1, \ldots, d$ and each pair of d-tuples
$$\begin{aligned} \underline{u} &:= (u_1, \ldots, u_{i-1}, u_i, u_{i+1}, \ldots, u_d) \in \mathcal{D}, \\ \underline{\widetilde{u}} &:= (u_1, \ldots, u_{i-1}, \widetilde{u}_i, u_{i+1}, \ldots, u_d) \in \mathcal{D}, \end{aligned}$$
such that $u_i \leq \widetilde{u}_i$, it follows that $F(\underline{u}) \leq F(\underline{\widetilde{u}})$.
- satisfies the *Lipschitz condition* if for given d–tuples $(u_1, \ldots, u_d)$ and $(v_1, \ldots, v_d)$ in $\mathcal{D}$, it holds that
$$|F(u_1, \ldots, u_d) - F(v_1, \ldots, v_d)| \leq \sum_{i=1}^d |u_i - v_i|.$$

If $\mathcal{D} = [0,1]^d$ and F satisfies the boundary, the monotonicity and the Lipschitz conditions, then F is called a *d-variate quasi-copula*. We will omit the dimension d when it is clear from the context and write quasi-copula for short.

Let $\mathcal{B} := \prod_{j=1}^d [x_j, y_j]$, where $0 \leq x_j \leq y_j \leq 1$ for each j. Next we recall the definition of a k-dimensional section of a function $F : \mathcal{B} \to [0,1]$ (see [5, Definition 5]). For any $\underline{a} := (a_1, \ldots, a_d) \in \mathcal{B}$ and a set of indices $A \subseteq \{1, 2, \ldots, d\}$ with $0 < \operatorname{card} A = k \leq d$, the *$k$-dimensional slice* (or *k-slice*) $\mathcal{B}_{\underline{a},A}$ of $\mathcal{B}$ with fixed values given by $\underline{a}$ in the positions determined by A, is defined by

$$\mathcal{B}_{\underline{a},A} := \{(z_1, \ldots, z_d) \colon z_j = a_j \text{ if } j \notin A \text{ and } z_j \in [x_j, y_j] \text{ if } j \in A.\}$$

In particular, if $\underline{a}$ is a vertex of $\mathcal{B}$, then $\mathcal{B}_{\underline{a},A}$ is called a *k-dimensional face* (or *k-face*) of $\mathcal{B}$. A *k-dimensional section* (or *k-section*) of F with fixed values given by $\underline{a}$ in the positions determined by A, is the function

$$F_{\underline{a},A} : B_{\underline{a},A} \to [0,1], \tag{2.1}$$

defined by

$$F_{\underline{a},A}(\underline{z}) := F(\underline{w}), \quad \text{where} \quad w_j = \begin{cases} z_j, & \text{if } j \in A, \\ a_j, & \text{if } j \notin A. \end{cases}$$

We say that a function $F : \mathcal{B} \to [0,1]$ is *k-dimensionally-increasing* (or *k-increasing*), with $k \in \{1, \ldots, d\}$, if any of its k-sections is k-increasing,

We denote by $\mathrm{MNV}(d,k)$ the maximal negative volume $V_Q(\mathcal{B})$, i.e., the largest in absolute value among negative ones, of some d-box $\mathcal{B}$, taken over all d-boxes $\prod_{i=1}^d [a_i, b_i] \subseteq [0,1]^d$ and over all k-increasing d-variate quasi-copulas Q. The following result provides an explicit formula for $\mathrm{MNV}(d,k)$.

Theorem 1. *Assume the notation above. Let $d, k \in \mathbb{N}$ with $2 \leq k \leq d$. Then*

$$\mathrm{MNV}(d,k) = \begin{cases} 0, & \text{if } k = d, \\ -\frac{\binom{d-1}{r_0}}{\binom{d-1}{k-1}}, & \text{if } k < d, \end{cases}$$

where

$$r_0 = \begin{cases} d-k-1, & d-k \leq \left\lfloor \frac{d-1}{2} \right\rfloor, \\ \left\lfloor \frac{d-1}{2} \right\rfloor + (-1)^d \dfrac{1-(-1)^\Delta}{2}, & d-k > \left\lfloor \frac{d-1}{2} \right\rfloor, \end{cases}$$

with $\Delta := d-k-1-\left\lfloor \frac{d-1}{2} \right\rfloor$.

For $k < d$, one realization of a k-increasing d-variate quasi-copula Q and a d-box $\mathcal{B}$ such that $V_Q(\mathcal{B}) = \mathrm{MNV}(d,k)$ is given by

$$\mathcal{B} = [a,1]^d, \qquad a = \frac{k-1}{d-1-r_0}, \qquad Q(z) = q_{d-S(z)}, \quad z \in \mathrm{Vert}(\mathcal{B}),$$

where

$$q_i = \frac{\binom{i-r_0-1}{k-1}}{\binom{d-1-r_0}{k-1}}, \qquad i = 0, \ldots, d,$$

with the convention $\binom{n}{m} = 0$ for $n < m$. This realization of Q on $\{a,1\}^d$ extends to a quasi-copula $Q : [0,1]^d \to [0,1]$ by Proposition 1.

We denote by $\mathrm{MPV}(d,k)$ the maximal positive volume $V_Q(\mathcal{B})$ of some d-box $\mathcal{B}$, taken over all d-boxes $\prod_{i=1}^d [a_i, b_i] \subseteq [0,1]^d$ and over all k-increasing d-variate quasi-copulas Q. The following result provides an explicit formula for $\mathrm{MPV}(d,k)$.

Theorem 2. *Assume the notation above. Let $d, k \in \mathbb{N}$ with $2 \leq k \leq d$. Then*

$$\mathrm{MPV}(d,k) = \frac{\binom{d-1}{r_0}}{\binom{d-1}{k-1}},$$

where

$$r_0 = \begin{cases} d-k, & d-k \leq \left\lfloor \frac{d-1}{2} \right\rfloor, \\ \left\lfloor \frac{d-1}{2} \right\rfloor + (-1)^d \dfrac{1+(-1)^\Delta}{2}, & d-k > \left\lfloor \frac{d-1}{2} \right\rfloor, \end{cases}$$

with $\Delta := d-k-\left\lfloor \frac{d-1}{2} \right\rfloor$.

One realization of a k-increasing d-variate quasi-copula Q and a d-box $\mathcal{B}$ such that $V_Q(\mathcal{B}) = \mathrm{MPV}(d,k)$ is given by

$$\mathcal{B} = [a,1]^d, \qquad a = \frac{k-1}{d-1-r_0}, \qquad Q(z) = q_{d-S(z)}, \quad z \in \mathrm{Vert}(\mathcal{B}),$$

where

$$q_i = \frac{\binom{i-r_0-1}{k-1}}{\binom{d-1-r_0}{k-1}}, \qquad i = 0,\dots,d,$$

with the convention $\binom{n}{m} = 0$ for $n < m$. This realization of Q on $\{a,1\}^d$ extends to a quasi-copula $Q : [0,1]^d \to [0,1]$ by Proposition 1.

Remark 1. (1) *Choice of the index r_0 in Theorems 1 and 2:* In both the negative and positive cases, the index r_0 is chosen so as to maximize the binomial coefficient $\binom{d-1}{r}$ under the constraint $0 \le r \le d-k$, together with a parity condition that arises from the sign pattern in the expressions defining $\mathrm{MNV}(d,k)$ and $\mathrm{MPV}(d,k)$. More precisely, the admissible parity is $r \equiv d-k-1 \pmod 2$ in the case of $\mathrm{MNV}(d,k)$ and $r \equiv d-k \pmod 2$ in the case of $\mathrm{MPV}(d,k)$. Since the sequence $r \mapsto \binom{d-1}{r}$ is unimodal with its maximum attained at $r = \lfloor \frac{d-1}{2} \rfloor$, the optimal choice of r_0 is the admissible index of the required parity that is closest to this peak, unless the truncation $r \le d-k$ is active. In the truncated regime, the maximum is attained at the largest admissible index. The explicit formulas for r_0 in both theorems encode exactly these considerations.

(2) *Simplified extremal values in the truncated regime:* Assume that $d-k \le \lfloor \frac{d-1}{2} \rfloor$. In this truncated regime, the explicit formulas for the extremal volumes simplify to

$$\mathrm{MNV}(d,k) = -\frac{d-k}{k}, \qquad \mathrm{MPV}(d,k) = 1.$$

Here the extremal values are attained at the boundary of the admissible index range, and no parity adjustment near the peak of the binomial coefficients is required.

Since $\mathrm{MPV}(d,k) = 1$ throughout this regime, an analysis based solely on the maximal positive volume might suggest that, for sufficiently large k, the class of k-increasing d-variate quasi-copulas coincides with the class of d-increasing d-quasi-copulas, which is precisely the class of copulas. However, this conclusion is false. The maximal negative volume $\mathrm{MNV}(d,k)$ provides a genuine distinction between these classes: even in the regime where $\mathrm{MPV}(d,k) = 1$, the values of $\mathrm{MNV}(d,k)$ do not coincide with those corresponding to copulas, which is 0. Thus, while maximal positive volumes alone fail to distinguish between the two classes, the maximal negative volumes show that the class of k-increasing quasi-copulas remains strictly larger than the class of copulas.

(3) *Relation between $\mathrm{MNV}(d,k)$ and $\mathrm{MPV}(d,k)$:* For $k < d$, the explicit formulas show that $\mathrm{MNV}(d,k)$ and $\mathrm{MPV}(d,k)$ differ only by the choice of the admissible parity class in the maximization of $\binom{d-1}{r}$. In particular, both quantities are given by ratios of binomial coefficients of the form $\frac{\binom{d-1}{r_0}}{\binom{d-1}{k-1}}$, with r_0 chosen as close as possible to $\lfloor \frac{d-1}{2} \rfloor$ subject to the respective parity constraint. Consequently, the magnitudes of $\mathrm{MNV}(d,k)$ and $\mathrm{MPV}(d,k)$ are comparable, and the difference between them reflects only the alternating sign structure inherent in the definition of the volume functional.

We may therefore consider the maximal signed volume as the pair

$$\mathrm{MSV}(d,k) := \big(\mathrm{MNV}(d,k),\ \mathrm{MPV}(d,k)\big);$$

note that $\mathrm{MSV}(2,2)$ may be viewed as a possible statistical counterpart of the ARV introduced in [3].

In Table 1 and 2 we give the values of $\mathrm{MNV}(d,k)$ and $\mathrm{MPV}(d,k)$ for dimensions up d to 15[1]. The cases $k \ge 2$ are computed according to Theorems 1 and 2 above, while the case $k = 1$ was studied in our previous work [11, Theorems 1 and 2].

[1]The numerical analysis in arithmetic over $\mathbb{Q}$ was performed using the software tool *Mathematica* [13]. The source code is available at `https://github.com/ZalarA/Quasi-copulas-k-increasing-extreme-volumes`.

TABLE 1. Explicit values of $\mathrm{MNV}(d,k)$ for $2 \le d \le 15$ and $1 \le k \le d$.

$k \backslash d$	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	$-\frac{1}{3}$	$-\frac{4}{5}$	$-\frac{9}{7}$	$-\frac{32}{13}$	$-\frac{75}{16}$	$-\frac{19}{2}$	$-\frac{55}{3}$	-37	$-\frac{209}{3}$	$-\frac{251}{2}$	$-\frac{791}{3}$	$-\frac{1583}{3}$	$-\frac{3002}{3}$	$-\frac{7435}{4}$
2	0	$-\frac{1}{2}$	-1	$-\frac{3}{2}$	-2	$-\frac{5}{2}$	-5	$-\frac{35}{4}$	-14	-21	-42	-77	-132	$-\frac{429}{2}$
3		0	$-\frac{1}{3}$	$-\frac{2}{3}$	-1	$-\frac{4}{3}$	$-\frac{5}{3}$	-2	$-\frac{7}{2}$	$-\frac{28}{5}$	$-\frac{42}{5}$	-12	-22	$-\frac{264}{7}$
4			0	$-\frac{1}{4}$	$-\frac{1}{2}$	$-\frac{3}{4}$	-1	$-\frac{5}{4}$	$-\frac{3}{2}$	$-\frac{7}{4}$	$-\frac{14}{5}$	$-\frac{21}{5}$	-6	$-\frac{33}{4}$
5				0	$-\frac{1}{5}$	$-\frac{2}{5}$	$-\frac{3}{5}$	$-\frac{4}{5}$	-1	$-\frac{6}{5}$	$-\frac{7}{5}$	$-\frac{8}{5}$	$-\frac{12}{5}$	$-\frac{24}{7}$
6					0	$-\frac{1}{6}$	$-\frac{1}{3}$	$-\frac{1}{2}$	$-\frac{2}{3}$	$-\frac{5}{6}$	-1	$-\frac{7}{6}$	$-\frac{4}{3}$	$-\frac{3}{2}$
7						0	$-\frac{1}{7}$	$-\frac{2}{7}$	$-\frac{3}{7}$	$-\frac{4}{7}$	$-\frac{5}{7}$	$-\frac{6}{7}$	-1	$-\frac{8}{7}$
8							0	$-\frac{1}{8}$	$-\frac{1}{4}$	$-\frac{3}{8}$	$-\frac{1}{2}$	$-\frac{5}{8}$	$-\frac{3}{4}$	$-\frac{7}{8}$
9								0	$-\frac{1}{9}$	$-\frac{2}{9}$	$-\frac{1}{3}$	$-\frac{4}{9}$	$-\frac{5}{9}$	$-\frac{2}{3}$
10									0	$-\frac{1}{10}$	$-\frac{1}{5}$	$-\frac{3}{10}$	$-\frac{2}{5}$	$-\frac{1}{2}$
11										0	$-\frac{1}{11}$	$-\frac{2}{11}$	$-\frac{3}{11}$	$-\frac{4}{11}$
12											0	$-\frac{1}{12}$	$-\frac{1}{6}$	$-\frac{1}{4}$
13												0	$-\frac{1}{13}$	$-\frac{2}{13}$
14													0	$-\frac{1}{14}$
15														0

Table 2. Explicit values of $\mathrm{MPV}(d,k)$ for $2 \le d \le 15$ and $1 \le k \le d$.

$k \backslash d$	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	1	1	2	$\frac{7}{2}$	$\frac{11}{2}$	$\frac{31}{3}$	19	$\frac{71}{2}$	$\frac{211}{3}$	$\frac{421}{3}$	$\frac{793}{3}$	$\frac{1915}{4}$	$\frac{3004}{3}$	$\frac{6007}{3}$
2	1	1	1	1	2	$\frac{10}{3}$	5	7	14	$\frac{126}{5}$	42	66	132	$\frac{1716}{7}$
3		1	1	1	1	1	$\frac{5}{3}$	$\frac{5}{2}$	$\frac{7}{2}$	$\frac{14}{3}$	$\frac{42}{5}$	14	22	33
4			1	1	1	1	1	1	$\frac{3}{2}$	$\frac{21}{10}$	$\frac{14}{5}$	$\frac{18}{5}$	6	$\frac{66}{7}$
5				1	1	1	1	1	1	1	$\frac{7}{5}$	$\frac{28}{15}$	$\frac{12}{5}$	3
6					1	1	1	1	1	1	1	1	$\frac{4}{3}$	$\frac{12}{7}$
7						1	1	1	1	1	1	1	1	1
8							1	1	1	1	1	1	1	1
9								1	1	1	1	1	1	1
10									1	1	1	1	1	1
11										1	1	1	1	1
12											1	1	1	1
13												1	1	1
14													1	1
15														1

3. Towards the proof of Theorems 1 and 2

In this section, we formulate the maximal volume problems as linear programs (see Proposition 1). We then simplify these programs in two steps: first, by applying the symmetrization trick (see Corollary 1), and second, by introducing new sets of variables (see Proposition 3). We note that the symmetrization trick played a crucial role in solving the maximal volume problems for quasi-copulas in our previous work [11], whereas the introduction of new sets of variables is the main novelty of the present paper, as it substantially reduces the complexity of the resulting linear programs. Finally, we derive the corresponding dual linear programs (see Proposition 4), which will be used in the next section to prove Theorems 1 and 2.

Fix $d \in \mathbb{N}$. For multi-indices

$$\mathbb{I} = (\mathbb{I}_1, \ldots, \mathbb{I}_d) \in \{0,1\}^d \quad \text{and} \quad \mathbb{J} = (\mathbb{J}_1, \ldots, \mathbb{J}_d) \in \{0,1\}^d$$

let

$$\mathbb{J} - \mathbb{I} = (\mathbb{J}_1 - \mathbb{I}_1, \ldots, \mathbb{J}_d - \mathbb{I}_d) \in \{-1,0,1\}^d$$

stand for their usual coordinate-wise difference. Let $\mathbb{E}^{(\ell)}$ stand for the multi-index with the only non-zero coordinate the ℓ–th one, which is equal to 1. For each $\ell = 1, \ldots, d$ we define a relation on $\{0,1\}^d$ by

$$\mathbb{I} \prec_\ell \mathbb{J} \quad \Leftrightarrow \quad \mathbb{J} - \mathbb{I} = \mathbb{E}^{(\ell)}.$$

We write

$$\mathbb{I} \prec \mathbb{J} \quad \Leftrightarrow \quad \mathbb{I} \prec_\ell \mathbb{J} \quad \text{for some } \ell \in \{1, 2, \ldots, d\}.$$

For a point $\underline{x} = (x_1, \ldots, x_d) \in \mathbb{R}^d$ we define the functions

$$G_d : \mathbb{R}^d \to \mathbb{R}, \quad G_d(\underline{x}) := \sum_{i=1}^{d} x_i - d + 1,$$
$$H_d : \mathbb{R}^d \to \mathbb{R}, \quad H_d(\underline{x}) := \min\{x_1, x_2, \ldots, x_d\}.$$

Let Q be a quasi-copula and $\mathcal{B} = \prod_{i=1}^d [a_i, b_i] \subseteq [0,1]^d$ a d-box with $a_i \le b_i$ for each i. We will use multi-indices of the form $\mathbb{I} := (\mathbb{I}_1, \mathbb{I}_2 \dots \mathbb{I}_d) \in \{0,1\}^d$ to index 2^d vertices of $\mathcal{B}$. Let $\|\mathbb{I}\|_1 := \sum_{j=1}^d \mathbb{I}_j$ be the 1-norm of $\mathbb{I}$. We write

$$x_{\mathbb{I}} := ((x_{\mathbb{I}})_1, \dots, (x_{\mathbb{I}})_d)$$

to denote the vertex with coordinates

$$(x_{\mathbb{I}})_k = \begin{cases} a_k, & \text{if } \mathbb{I}_k = 0, \\ b_k, & \text{if } \mathbb{I}_k = 1. \end{cases} \tag{3.1}$$

Let us denote the value of Q in the point $x_{\mathbb{I}}$ by

$$q_{\mathbb{I}} := Q(x_{\mathbb{I}}).$$

We write $\operatorname{sign}(\mathbb{I}) := (-1)^{d - \|\mathbb{I}\|_1}$. In the notation above, the Q-volume of $\mathcal{B}$ is equal to

$$V_Q(\mathcal{B}) = \sum_{\mathbb{I} \in \{0,1\}^d} \operatorname{sign}(\mathbb{I}) \cdot q_{\mathbb{I}}.$$

Let F be a face of the unit d-cube $[0,1]^d$. We denote with

$$\mathbb{I}(F) := \{\mathbb{I} \colon \mathbb{I} \in \operatorname{Vert}(F)\} \subseteq \{0,1\}^d$$

the set of all multi-indices, corresponding to the vertices of the face F. Let

$$\mathbb{I}_{\max}(F) := (\max_{\mathbb{I} \in \mathbb{I}(F)} \mathbb{I}_1, \max_{\mathbb{I} \in \mathbb{I}(F)} \mathbb{I}_2, \dots, \max_{\mathbb{I} \in \mathbb{I}(F)} \mathbb{I}_d)$$

be the vertex of F such that each of its coordinates is the largest among all vertices. We define

$$\operatorname{sign}_F(\mathbb{I}) := (-1)^{\|\mathbb{I}_{\max}(F) - \mathbb{I}\|_1}.$$

We will prove Theorems 1 and 2 in several steps using linear programming as the main tool.

Proposition 1. *For any $k \in \{1, 2, \dots, d\}$ define the following linear program*

$$\begin{aligned} \min_{\substack{a_1,\dots,a_d,\\ b_1,\dots,b_d,\\ q_{\mathbb{I}} \text{ for } \mathbb{I} \in \{0,1\}^d}} \quad & \sum_{\mathbb{I} \in \{0,1\}^d} \operatorname{sign}(\mathbb{I}) q_{\mathbb{I}}, \\ \text{subject to} \quad & 0 \le a_i < b_i \le 1 \quad \text{for all } i = 1, 2, \dots, d, \\ & q_{\mathbb{J}} - q_{\mathbb{I}} \le b_\ell - a_\ell \quad \text{for all } \ell = 1, 2, \dots, d \text{ and all } \mathbb{I} \prec_\ell \mathbb{J}, \\ & 0 \le \sum_{\mathbb{J} \in \mathbb{I}(F)} \operatorname{sign}_F(\mathbb{J}) \cdot q_{\mathbb{J}} \quad \text{for all } j\text{-dimensional faces } F \text{ of } [0,1]^d \\ & \qquad\qquad\qquad \text{and for all } j = 1, \dots, k, \\ & \max\{0, G_d(x_{\mathbb{I}})\} \le q_{\mathbb{I}} \le H_d(x_{\mathbb{I}}) \quad \text{for all } \mathbb{I} \in \{0,1\}^d. \end{aligned} \tag{3.2}$$

Let $\mathbb{I}^{(1)}, \dots, \mathbb{I}^{(2^d)}$ be some order of all multi-indices $\mathbb{I} \in \{0,1\}^d$. If there exists an optimal solution

$$(a_1^*, \dots, a_d^*, b_1^*, \dots, b_d^*, q_{\mathbb{I}^{(1)}}, \dots, q_{\mathbb{I}^{(2^d)}})$$

to (3.2)*, which satisfies*

$$b_1^* = \ldots = b_d^* = 1, \tag{3.3}$$

then the optimal value of (3.2) *is the maximal negative volume of a d-box, taken over all d-boxes $\prod_{i=1}^d [a_i, b_i] \subseteq [0,1]^d$ and over all k-increasing d-variate quasi-copulas Q.*

Moreover, if there exists an optimal solution to (3.2) *where* min *is replaced with* max*, which satisfies* (3.3)*, then the optimal value of* (3.2) *is the maximal positive volume of a d-box, taken over all d-boxes $\prod_{i=1}^d [a_i, b_i] \subseteq [0,1]^d$ and over all k-increasing d-variate quasi-copulas Q..*

Remark 2. (1) The condition (3.3) in Proposition 1 is necessary, as it ensures that the optimal solution to (3.2) indeed extends to a quasi-copula. Without this condition, it may happen that the values at points with some coordinates equal to b_ℓ are too small, preventing the function from increasing to 1 at the point $(1, \dots, 1)$. In our previous work [10], where we studied maximal signed volumes over all quasi-copulas (that is, the case $k = 1$ in (3.2)), we had to treat separately the situations in which the optimal solution of (3.2) did not satisfy this requirement. More precisely, we solved an alternative linear program (see [10, (3.4) on p. 15]), where the underlying grid was expanded from a 2^d-point grid $\prod_{i=1}^d \{a_i, b_i\}$ to a 3^d-point grid $\prod_{i=1}^d \{a_i, b_i, 1\}$. We then imposed additional Lipschitz conditions to ensure that the solution indeed extends to a quasi-copula.

(2) We also note that, in the proof of Proposition 1, the role of the condition (3.3) is not explicitly apparent. Nevertheless, it is implicitly required in the first sentence of the last paragraph of the proof, where we invoke [10, Theorem 2.1]. As discussed in the first part of this remark, without this condition the solution constructed in the proof of [10, Theorem 2.1] may fail to extend to a quasi-copula, due to insufficient values at points with at least one coordinate equal to b_ℓ.

In the proof of Proposition 1 we will use the fact that the k-increasing property of a multilinear function on a d-box follows from the k-increasing property on all k-faces of the box (see Lemma 1 below).

Lemma 1. *Let F be a multilinear function on a d-box $\mathcal{B} = \prod_{i=1}^d [x_i, y_i] \subseteq [0,1]^d$. The following statements are equivalent:*

(1) F is k-increasing.

(2) F is k-increasing on each k-face of $\mathcal{B}$.

Proof. The nontrivial implication is (2) $\Rightarrow$ (1). We will use induction on ℓ to show that F is k-increasing on each ℓ-dimensional face of $\mathcal{B}$ for $\ell = k, k+1, \ldots, d$. For $\ell = d$ we get that F is k-increasing on the whole box $\mathcal{B}$, proving (1). The base of induction is $\ell = k$ and holds by the assumption of (2). Assume now that F is k-increasing on each ℓ-face for some ℓ with $\ell \leq d-1$ and prove that it is k-increasing on each $(\ell+1)$-face. Let $\mathcal{B}_{\underline{u},A}$ be an arbitrary $(\ell+1)$-face of $\mathcal{B}$, where $\underline{u} \in \operatorname{Vert}\mathcal{B}$ and $A \subseteq \{1, 2, \ldots, d\}$ with $\operatorname{card} A = \ell + 1$. Let $\underline{z} \in \mathcal{B}_{\underline{u},A}$ and $(\mathcal{B}_{\underline{u},A})_{\underline{z},B}$ be a ℓ-section of $\mathcal{B}_{\underline{u},A}$, where $\underline{z} \in \mathcal{B}_{\underline{u},A}$ and $B \subset A$ with $\operatorname{card} B = \ell$. Let $\{i_0\} := A \setminus B$. Without loss of generality we can assume that $\underline{u}_{i_0} = x_{u_0}$. Then $z_{i_0} = \lambda x_{i_0} + (1-\lambda) y_{i_0}$ for some $\lambda \in [0,1]$ and thus the l section $(\mathcal{B}_{\underline{u},A})_{\underline{z},B}$ can be written as a convex combination of two parallel l sub-faces $\mathcal{B}_{\underline{u},B}$ and $\mathcal{B}_{\underline{v},B}$

$$(\mathcal{B}_{\underline{u},A})_{\underline{z},B} = \lambda \mathcal{B}_{\underline{u},B} + (1-\lambda)\mathcal{B}_{\underline{v},B}, \tag{3.4}$$

where $\underline{v}$ is a neighbor vertex of $\underline{u}$ that only differs in coordinate i_0

$$v_j = \begin{cases} u_j & j \neq i_0, \\ y_{i_0}, & j = i_0. \end{cases}$$

By the induction hypothesis, F is k-increasing on $\mathcal{B}_{\underline{u},B}$ and $\mathcal{B}_{\underline{v},B}$. By the multilinearity of F and by (3.4), F is k-increasing on $\mathcal{B}_{\underline{a},A}$. This completes the induction step and concludes the proof of the lemma. $\square$

Now we can prove Proposition 1.

Proof of Proposition 1. The linear program (3.2) is obtained by extending the linear program for general quasi-copulas introduced in [11, Proposition 2] with additional constraints enforcing the k-increasing property. By [5, Lemma 2], any k-increasing quasi-copula is also j-increasing for every $j \leq k$. Nevertheless, in the linear program it is necessary to impose the j-increasing constraints explicitly for all $j \leq k$. Otherwise, the resulting quasi-copula would satisfy the k-increasing property only on the box $\prod_{i=1}^k [a_i, b_i]$, which is insufficient to invoke [5, Lemma 2].

It remains to verify that the solution of the linear program (3.2) can be extended to a k-increasing quasi-copula.

Let points $x_{\mathbb{I}} \in \prod_{i=1}^d \{a_i, 1\} =: \mathcal{D}$ be defined by (3.1) and real numbers $q_{\mathbb{I}}$ for all $\mathbb{I} \in \{0,1\}^d$, satisfy conditions (3.2). Let $\mathcal{L}_i$ denote the $(d-1)$–dimensional faces of $[0,1]^d$ containing $(0, \ldots, 0)$, i.e.,

$$\mathcal{L}_i = \{(x_1, \ldots, x_{i-1}, 0, x_{i+1}, \ldots, x_d) \colon x_i \in [0,1]\}.$$

Let

$$\mathcal{D}^{(\mathrm{ext})} := \mathcal{D} \cup \Big(\cup_{i=1}^d \mathcal{L}_i \Big)$$

and define

$$Q : \mathcal{D}^{(\mathrm{ext})} \to \mathbb{R}$$

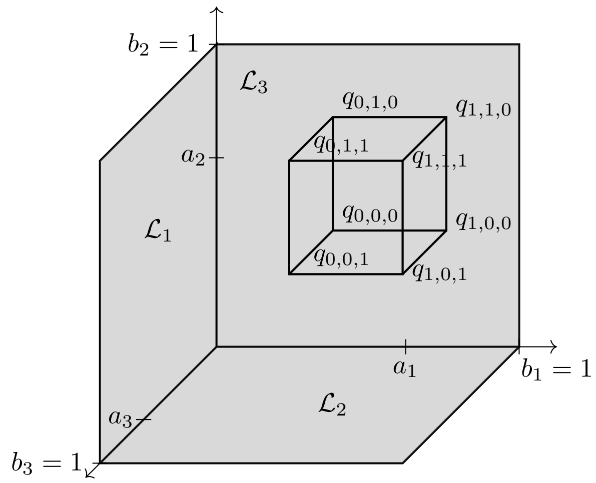

FIGURE 1. To construct a k-increasing 3-quasi-copula Q given the values $q_{\mathbb{I}}$ at $x_{\mathbb{I}} \in \mathcal{D} := \prod_{i=1}^{3}\{a_i, 1\}$, we first define it to be 0 on all 2-dimensional faces $\mathcal{L}_1$, $\mathcal{L}_2$, $\mathcal{L}_3$ and prove that the extension indeed meets the requirements of a k-increasing quasi-copula.

by

$$Q(x_1, \ldots, x_d) = \begin{cases} q_{\mathbb{I}}, & \text{if } (x_1, \ldots, x_d) = x_{\mathbb{I}} \text{ for some } \mathbb{I} \in \{0,1\}^d, \\ 0, & \text{if } (x_1, \ldots, x_d) \in \mathcal{L}_i \text{ for some } i \in \{1, \ldots, d\}. \end{cases}$$

We subdivide the box $[0,1]^d$ into 2^d smaller d-boxes

$$\mathcal{B}_{\mathbb{I}} = \prod_{j=1}^{d} \delta_j(\mathbb{I}) \tag{3.5}$$

for $\mathbb{I} = (\mathbb{I}_1, \ldots, \mathbb{I}_d) \in \{0,1\}^d$, where

$$\delta_j(\mathbb{I}) = \begin{cases} [0, a_j], & \text{if } \mathbb{I}_j = 0, \\ [a_j, 1], & \text{if } \mathbb{I}_j = 1. \end{cases}$$

In particular,

$$\mathcal{B}_{(0,0\ldots0)} = \prod_{k=1}^{d}[0, a_k], \quad \mathcal{B}_{(1,0\ldots0)} = [a_1, 1] \times \prod_{k=2}^{d}[0, a_k], \ldots, \quad \mathcal{B}_{(1,1\ldots1)} = \prod_{k=1}^{d}[a_k, 1].$$

Note that the Q-volume $V_Q(\mathcal{B}_{\mathbb{I}})$ of each box $\mathcal{B}_{\mathbb{I}}$ is determined by the value of Q on the vertices of $\mathcal{B}_{\mathbb{I}}$. For each $\mathbb{I} \in \{0,1\}^d$ we define a constant function

$$\rho_{\mathbb{I}} : \mathcal{B}_{\mathbb{I}} \to \mathbb{R}, \quad \rho_{\mathbb{I}} := \frac{V_Q(\mathcal{B}_{\mathbb{I}})}{\prod_{j=1}^{d} V(\delta_j(\mathbb{I}))},$$

where $V([a,b]) = |b-a|$ is the length of the interval $[a,b]$. Let us define a piecewise constant function

$$\rho : [0,1]^d \to \mathbb{R}, \quad \rho(\underline{x}) := \begin{cases} \rho_{\mathbb{I}}, & \text{if } \underline{x} \in \operatorname{int}(\mathcal{B}_{\mathbb{I}}) \text{ for some } \mathbb{I} \in \{0,1\}^d, \\ 0, & \text{otherwise}, \end{cases}$$

where $\operatorname{int}(A)$ stands for the topological interior of the set A in the usual Euclidean topology. We will prove that a function $Q : [0,1]^d \to \mathbb{R}$, defined by

$$Q(x_1, \ldots, x_d) = \int_0^{x_1} \cdots \int_0^{x_d} \rho(t_1, \ldots, t_d)\, dt_1 \cdots dt_d. \tag{3.6}$$

is a k–increasing quasi-copula satisfying the statement of the proposition.

By the same proof as in [10, Theorem 2.1], it follows that Q is a quasi-copula. It remains to prove that it is k-increasing. By Lemma 1, it suffices to show that Q is k-increasing on each k-face of each box $\mathcal{B}_{\mathbb{I}}$ for $\mathbb{I} \in \{0,1\}^d$. By construction, this holds for $\mathbb{I}_{\text{main}} = (1, 1, \ldots, 1)$. Now consider $\mathbb{I} \in \{0,1\}^d \setminus \{\mathbb{I}_{\text{main}}\}$. If the face has a constant 0 in some component, then Q is identically 0 on the entire face, so Q is k-increasing. Otherwise, every constant component is equal to a_j or 1, where $a_j \neq 0$. Let $(\mathcal{B}_{\mathbb{I}})_{\underline{u},U}$ be a k-face of $\mathcal{B}_{\mathbb{I}}$, where $\underline{u} = (u_1, \ldots, u_d) \in \operatorname{Vert} \mathcal{B}_{\mathbb{I}}$, $U \subseteq \{1, \ldots, d\}$ and $\operatorname{card} U = k$. By relabeling the indices we may assume that $U=\{1,\ldots,k\}$, $\mathbb{I}_1=\ldots=\mathbb{I}_\ell=0$ and $\mathbb{I}_{\ell+1}=\ldots=\mathbb{I}_k=1$, where $0\leq\ell\leq k$. Namely, $(\mathcal{B}_{\mathbb{I}})_{\underline{u},U}=\prod_{i=1}^{\ell}[0,a_i]\times\prod_{i=\ell+1}^{k}[a_i,1]\times\prod_{i=k+1}^{d}\{u_i\}$. Let $\mathcal{B}=\prod_{i=1}^{k}[c_i,d_i]$ be a k-box in $(\mathcal{B}_{\mathbb{I}})_{\underline{u},U}$ and define $\mathcal{B}^{(j)}:=\prod_{i=1}^{j}[0,a_i]\times\prod_{i=j+1}^{k}[c_i,d_i]$ for $j=1,\ldots,\ell$. By the multilinearity of Q in $\mathcal{B}_{\mathbb{I}}$, we have that

$$V_{Q_{\underline{u},U}}(\mathcal{B})=\frac{(d_1-c_1)}{a_1}V_{Q_{\underline{u},U}}(\mathcal{B}^{(1)})=\frac{(d_1-c_1)}{a_1}\frac{(d_2-c_2)}{a_2}V_{Q_{\underline{u},U}}(\mathcal{B}^{(2)})=\ldots=\Big(\prod_{i=1}^{\ell}\frac{(d_i-c_i)}{a_i}\Big)V_{Q_{\underline{u},U}}(\mathcal{B}^{(\ell)}).$$

Note that

$$V_{Q_{\underline{u},U}}(\mathcal{B}^{(\ell)})=V_{Q_{\underline{w},W}}\Big(\prod_{i=\ell+1}^{k}[c_i,d_i]\Big), \tag{3.7}$$

where $\underline{w}=(a_1,\ldots,a_k,u_{k+1},\ldots,u_d)$ and $W=\{\ell+1,\ldots,k\}$. But $Q_{\underline{w},W}$ is a $(k-\ell)$-dimensional face of $\mathcal{B}_{\mathbb{I}_{\text{main}}}$, which is k-increasing by construction. Therefore the volume in (3.7) is positive, so Q is k-increasing on $\mathcal{B}_{\mathbb{I}}$. □

By the following proposition it suffices to consider symmetric solutions to the linear program (3.2).

Proposition 2 ([11, Proposition 3])**.** *Assume the notation from Proposition 1. There exists an optimal solution to the linear program* (3.2) *of the form*

$$\big(\underbrace{a^*,\ldots,a^*}_{d},\underbrace{b^*,\ldots,b^*}_{d},q^*_{\|\mathbb{I}^{(1)}\|_1},\ldots,q^*_{\|\mathbb{I}^{(2^d)}\|_1}\big) \tag{3.8}$$

for some $a^*,b^*,q_0^*,q_1^*,\ldots,q_d^*\in[0,1]$.

Analogously, replacing min *with* max *in* (3.2) *above, the same statement holds.*

Proof. The proof is the same as for [10, Proposition 3], but we include it for the sake of completeness. Let S_d be the set of all permutations of a d-element set $\{1,\ldots,d\}$. For $\Phi\in S_d$ and $\mathbb{I}^{(j)}:=(\mathbb{I}_1^{(j)},\ldots,\mathbb{I}_d^{(j)})\in\{0,1\}^d$, let $\Phi(\mathbb{I}^{(j)}):=\big(\mathbb{I}^{(j)}_{\Phi(1)},\ldots,\mathbb{I}^{(j)}_{\Phi(d)}\big)$. For every optimal solution $\big(a_1^*,\ldots,a_d^*,b_1^*,\ldots,b_d^*,q^*_{\mathbb{I}^{(1)}},\ldots,q^*_{\mathbb{I}^{(2^d)}}\big)$ to (3.2), $\big(a^*_{\Phi(1)},\ldots,a^*_{\Phi(d)},b^*_{\Phi(1)},\ldots,b^*_{\Phi(d)},q^*_{\Phi(\mathbb{I}^{(1)})},\ldots,q^*_{\Phi(\mathbb{I}^{(2^d)})}\big)$ is also an optimal solution to (3.2), whence an optimal solution to (3.2) of the form (3.8) is equal to

$$\frac{1}{d!}\sum_{\Phi\in S_d}\big(a^*_{\Phi(1)},\ldots,a^*_{\Phi(d)},b^*_{\Phi(1)},\ldots,b^*_{\Phi(d)},q^*_{\Phi(\mathbb{I}^{(1)})},\ldots,q^*_{\Phi(\mathbb{I}^{(2^d)})}\big).$$

This concludes the proof. □

An immediate corollary to Proposition 2 is the following.

Corollary 1. *The optimal value of the linear program* (3.2) *is equal to the optimal value of the linear program*

$$\begin{aligned}
\min_{a,b,q_0,q_1,\ldots,q_d}\quad & \sum_{i=0}^{d}(-1)^{d-i}\binom{d}{i}q_i,\\
\textit{subject to}\quad & 0\leq a<b\leq 1,\\
& q_i-q_{i-1}\leq b-a \quad \textit{for } i=1,2,\ldots,d,\\
& 0\leq\sum_{i=0}^{j}(-1)^{j-i}\binom{j}{i}q_{\ell-i} \quad \textit{for all } j\in 1,2,\ldots,k \textit{ and all } \ell=j,j+1,\ldots,d,\\
& \max\{0,(d-i)a+ib-d+1\}\leq q_i\leq a \quad \textit{for } i=0,1,\ldots,d-1,\\
& \max\{0,db-d+1\}\leq q_d\leq b.
\end{aligned} \tag{3.9}$$

Analogously, replacing min *with* max *in* (3.9) *above, the same statement holds.*

Remark 3. Corollary 1 differs from [11, Corollary 1] in that, for the third type of constraints, the index j ranges from 1 to k in the present paper, whereas it is fixed at $j=1$ in [11, Corollary 1]. This distinction reflects the fact that a fixed value of j yields conditions for being j-increasing.

It turns out that some of the constraints in (3.9) are redundant, while introducing new variables

$$\begin{aligned} &\delta_i^{(1)} := q_i - q_{i-1} \quad \text{for } i = 1, 2, \ldots, d, \\ &\delta_i^{(j)} := \delta_i^{(j-1)} - \delta_{i-1}^{(j-1)} \quad \text{for } j \in 2, 3, \ldots, k \text{ and } i = j, j+1, \ldots, d. \end{aligned} \tag{3.10}$$

further decreases the number of constraints. Let us define the numbers $\alpha_i^{(j)}, \beta_i^{(j)}, \gamma_i^{(j)}$ recursively by

$$\begin{aligned} \alpha_i^{(j)} &= \begin{cases} 1 & \text{for } j = 2 \text{ and } i = 2, \ldots, d, \\ \alpha_i^{(j-1)} & \text{for } j \geq 3 \text{ and } i = 2, \ldots, j-2, \\ \sum_{\ell=i}^{d} \alpha_\ell^{(j-1)} & \text{for } j \geq 3 \text{ and } i = j-1, \ldots, d, \end{cases} \\ \beta_i^{(j)} &= \begin{cases} d-i & \text{for } j = 2 \text{ and } i = 2, \ldots, d-1, \\ \beta_i^{(j-1)} & \text{for } j \geq 3 \text{ and } i = 2, \ldots, j-2, \\ \sum_{\ell=i}^{d-1} \beta_\ell^{(j-1)} & \text{for } j \geq 3 \text{ and } i = j-1, \ldots, d-1, \end{cases} \\ \gamma_i^{(j)} &= \begin{cases} d+1-i & \text{for } j = 2 \text{ and } i = 2, \ldots, d, \\ \gamma_i^{(j-1)} & \text{for } j \geq 3 \text{ and } i = 2, \ldots, j-2, \\ \sum_{\ell=i}^{d} \gamma_\ell^{(j-1)} & \text{for } j \geq 3 \text{ and } i = j-1, \ldots, d. \end{cases} \end{aligned} \tag{3.11}$$

Proposition 3. *Let $\alpha_i^{(j)}$, $\beta_i^{(j)}$ and $\gamma_i^{(j)}$ be as in* (3.11). *The optimal value of the linear program* (3.9) *is equal to the optimal value of the linear program*

$$\begin{aligned} \min_{\substack{a,b,q_0,\\ \delta_1^{(1)},\delta_2^{(2)},\ldots,\delta_{k-1}^{(k-1)},\\ \delta_k^{(k)},\delta_{k+1}^{(k)},\ldots,\delta_d^{(k)}}} \quad & \sum_{j=k}^{d} (-1)^{d+j} \binom{d-k}{j-k} \delta_j^{(k)}, \\ \textit{subject to} \quad & b \leq 1, \\ & a - b + \delta_1^{(1)} + \sum_{i=2}^{k-1} \alpha_i^{(k)} \delta_i^{(i)} + \sum_{i=k}^{d} \alpha_i^{(k)} \delta_i^{(k)} \leq 0, \\ & -a + q_0 + (d-1)\delta_1^{(1)} + \sum_{i=2}^{k-1} \beta_i^{(k)} \delta_i^{(i)} + \sum_{i=k}^{d-1} \beta_i^{(k)} \delta_i^{(k)} \leq 0, \\ & db - q_0 - d\delta_1^{(1)} - \sum_{i=2}^{k-1} \gamma_i^{(k)} \delta_i^{(i)} - \sum_{i=k}^{d} \gamma_i^{(k)} \delta_i^{(k)} \leq d-1, \\ & a \geq 0,\ b \geq 0,\ q_0 \geq 0,\ \delta_i^{(i)} \geq 0 \ \textit{for } i = 1, 2, \ldots, k-1, \\ & \delta_i^{(k)} \geq 0 \ \textit{for } i = k, k+1, \ldots, d. \end{aligned} \tag{3.12}$$

Analogously, replacing min *with* max *in* (3.12) *above, the same statement holds.*

Proof. We will use induction on k to prove Proposition 3.

We start by $k = 2$. By [11, Proposition 5], (3.9) is equivalent to

$$\begin{aligned}
&\min_{a,b,q_0,\delta_1^{(1)},\ldots,\delta_d^{(1)}} && \sum_{j=1}^{d}(-1)^{d+j}\binom{d-1}{j-1}\delta_j^{(1)},\\
&\text{subject to} && b \le 1,\\
& && \delta_i^{(1)} \le b-a \quad \text{for } i = 1,2,\ldots,d,\\
& && 0 \le q_i - 2q_{i-1} + q_{i-2} \quad \text{for } i = 2,3,\ldots,d,\\
& && q_0 + \sum_{i=1}^{d-1}\delta_i^{(1)} \le a\\
& && db - d + 1 \le q_0 + \sum_{i=1}^{d}\delta_i^{(1)}\\
& && a \ge 0,\ b \ge 0,\ q_0 \ge 0,\ \delta_i^{(1)} \ge 0 \quad \text{for } i = 1,2,\ldots,d.
\end{aligned} \tag{3.13}$$

Since $\delta_i^{(1)}, \delta_i^{(2)}$ are as in (3.10), we can replace the inequalities

$$0 \le q_i - 2q_{i-1} + q_{i-2} \quad \text{for } i = 2,3,\ldots,d$$

with the inequalities

$$0 \le \delta_i^{(2)} \quad \text{for } i = 2,3,\ldots,d. \tag{3.14}$$

Using (3.14), the constraints $\delta_i^{(1)} \ge 0$ and $\delta_i^{(1)} \le b-a$ for $i = 1,2,\ldots,d$ in (3.13) can be replaced by the constraints $\delta_1^{(1)} \ge 0$ and $\delta_d^{(1)} \le b-a$. Since $\delta_j^{(1)} = \delta_j^{(2)} + \delta_{j-1}^{(1)}$ for $j = 2,3,\ldots,d$, it follows inductively that

$$\delta_j^{(1)} = \sum_{l=2}^{j}\delta_l^{(2)} + \delta_1^{(1)} \quad \text{for } j = 2,3,\ldots,d. \tag{3.15}$$

Using (3.15) in the constraints of (3.13), it is easy to see that we get the constraints of (3.9) with $\alpha_i^{(2)}, \beta_i^{(2)}, \gamma_i^{(2)}$. It remains to show that the objective function of (3.9) becomes the objective function of (3.12) after substitutions (3.15). Indeed,

$$\begin{aligned}
\sum_{j=1}^{d}(-1)^{d-j}\binom{d-1}{j-1}\delta_j^{(1)} &= \sum_{j=1}^{d}(-1)^{d-j}\binom{d-1}{j-1}\Big(\delta_1^{(1)} + \sum_{\ell=2}^{j}\delta_\ell^{(2)}\Big)\\
&= \delta_1^{(1)}\underbrace{\sum_{j=1}^{d}(-1)^{d-j}\binom{d-1}{j-1}}_{0} + \sum_{\ell=2}^{d}\delta_\ell^{(2)}\underbrace{\Big(\sum_{k=\ell}^{d}(-1)^{d-k}\binom{d-1}{k-1}\Big)}_{(-1)^{d+\ell}\binom{d-2}{\ell-2}},
\end{aligned}$$

where in the last equality we used that for $0 \le n \le r$ we have

$$\sum_{k=n}^{r}(-1)^k\binom{r}{k} \underbrace{=}_{\sum_{k=0}^{r}(-1)^k\binom{r}{k}=0} -\sum_{k=0}^{n-1}(-1)^k\binom{r}{k} = (-1)^n\binom{r-1}{n-1}. \tag{3.16}$$

Assume know that the proposition holds for some k, $2 \le k \le d-1$ and prove it for $k+1$. By the induction hypothesis the optimal value of the linear program (3.9) is equal to the optimal value of the linear program

$$
\begin{aligned}
\min_{\substack{a,b,q_0,\\ \delta_1^{(1)},\delta_2^{(2)},\ldots,\delta_{k-1}^{(k-1)},\\ \delta_k^{(k)},\delta_{k+1}^{(k)},\ldots,\delta_d^{(k)}}} \quad & \sum_{j=k}^{d}(-1)^{d+j}\binom{d-k}{j-k}\delta_j^{(k)}, \\
\text{subject to} \quad & b\leq 1, \\
& a-b+\delta_1^{(1)}+\sum_{i=2}^{k-1}\alpha_i^{(k)}\delta_i^{(i)}+\sum_{i=k}^{d}\alpha_i^{(k)}\delta_i^{(k)}\leq 0, \\
& 0\leq \sum_{i=0}^{k+1}(-1)^{k+1-i}\binom{k+1}{i}q_{\ell-i} \quad \text{for all } \ell=k+1,k+2,\ldots,d, \\
& -a+q_0+(d-1)\delta_1^{(1)}+\sum_{i=2}^{k-1}\beta_i^{(k)}\delta_i^{(i)}+\sum_{i=k}^{d-1}\beta_i^{(k)}\delta_i^{(k)}\leq 0, \\
& db-q_0-d\delta_1^{(1)}-\sum_{i=2}^{k-1}\gamma_i^{(k)}\delta_i^{(i)}-\sum_{i=k}^{d}\gamma_i^{(k)}\delta_i^{(k)}\leq d-1, \\
& a\geq 0,\ b\geq 0,\ q_0\geq 0,\ \delta_i^{(i)}\geq 0 \ \text{ for } i=1,2,\ldots,k-1, \\
& \delta_i^{(k)}\geq 0 \ \text{ for } i=k,k+1,\ldots,d.
\end{aligned}
\tag{3.17}
$$

Since $\delta_i^{(j)}$ are as in (3.10), we can replace the inequalities

$$0\leq \sum_{i=0}^{k+1}(-1)^{k+1-i}\binom{k+1}{i}q_{\ell-i} \quad \text{for all } \ell=k+1,k+2,\ldots,d,$$

with the inequalities

$$0\leq \delta_\ell^{(k+1)} \quad \text{for } \ell=k+1,k+2,\ldots,d.$$

Since $\delta_j^{(k)}=\delta_j^{(k+1)}+\delta_{j-1}^{(k)}$ for $j=k+1,k+2,\ldots,d$, it follows inductively that

$$\delta_j^{(k)}=\sum_{\ell=k+1}^{j}\delta_\ell^{(k+1)}+\delta_k^{(k)} \quad \text{for } j=k+1,k+2,\ldots,d. \tag{3.18}$$

Using (3.18) in the constraints of (3.17), it is easy to see that the constraints become as in (3.12), where k is replaced by $k+1$. It remains to show that the objective function of (3.17) becomes the objective function of (3.12) after substitutions (3.18). Indeed,

$$
\begin{aligned}
\sum_{j=k}^{d}(-1)^{d+j}\binom{d-k}{j-k}\delta_j^{(k)} &= \sum_{j=k}^{d}(-1)^{d+j}\binom{d-k}{j-k}\Big(\delta_k^{(k)}+\sum_{\ell=k+1}^{j}\delta_\ell^{(k+1)}\Big) \\
&= \delta_k^{(k)}\underbrace{\sum_{j=k}^{d}(-1)^{d+j}\binom{d-k}{j-k}}_{0}+\sum_{\ell=k+1}^{d}\delta_\ell^{(k+1)}\underbrace{\Big(\sum_{j=\ell}^{d}(-1)^{d+j}\binom{d-k}{j-k}\Big)}_{(-1)^{d+\ell}\binom{d-k-1}{\ell-k-1}},
\end{aligned}
$$

where in the last equality we used (3.16). □

Using duality, the following proposition holds.

Proposition 4. *Let $\alpha_i^{(j)}$, $\beta_i^{(j)}$ and $\gamma_i^{(j)}$ be as in* (3.11). *The optimal value of the linear program* (3.12) *is equal to the optimal value of the linear program*

$$
\begin{aligned}
\max_{y_1,y_2,y_3,y_4} \quad & -y_1-(d-1)y_4,\\
\text{subject to} \quad & y_1-y_2+dy_4\geq 0,\\
& y_2-y_3\geq 0,\\
& y_3-y_4\geq 0,\\
& y_2+(d-1)y_3-dy_4\geq 0,\\
& \alpha_i^{(k)}y_2+\beta_i^{(k)}y_3-\gamma_i^{(k)}y_4\geq 0 \quad \text{for } i=2,3,\ldots,k-1,\\
& \alpha_i^{(k)}y_2+\beta_i^{(k)}y_3-\gamma_i^{(k)}y_4\geq (-1)^{d+1+i}\binom{d-k}{i-k} \quad \text{for } i=k,k+1,\ldots,d,\\
& y_1\geq 0,\ y_2\geq 0,\ y_3\geq 0,\ y_4\geq 0.
\end{aligned}
\tag{3.19}
$$

Analogously, the optimal value of the linear program (3.12) *with the objective function*

$$
\max_{\substack{a,b,q_0,\\ \delta_1^{(1)},\delta_2^{(2)},\ldots,\delta_{k-1}^{(k-1)},\\ \delta_k^{(k)},\delta_{k+1}^{(k)},\ldots,\delta_d^{(k)}}} \sum_{j=k}^{d}(-1)^{d+j}\binom{d-k}{j-k}\delta_j^{(k)},
$$

is equal to the optimal value of the linear program

$$
\begin{aligned}
\min_{y_1,y_2,y_3,y_4} \quad & y_1+(d-1)y_4,\\
\text{subject to} \quad & y_1-y_2+dy_4\geq 0,\\
& y_2-y_3\geq 0,\\
& y_3-y_4\geq 0,\\
& y_2+(d-1)y_3-dy_4\geq 0,\\
& \alpha_i^{(k)}y_2+\beta_i^{(k)}y_3-\gamma_i^{(k)}y_4\geq 0 \quad \text{for } i=2,3,\ldots,k-1,\\
& \alpha_i^{(k)}y_2+\beta_i^{(k)}y_3-\gamma_i^{(k)}y_4\geq (-1)^{d+i}\binom{d-k}{i-k} \quad \text{for } i=k,k+1,\ldots,d,\\
& y_1\geq 0,\ y_2\geq 0,\ y_3\geq 0,\ y_4\geq 0.
\end{aligned}
\tag{3.20}
$$

Proof. Note that (3.19) is the dual linear program to (3.12), whence the statement of Proposition 4 follows by the strong duality. □

4. Proofs of Theorems 1 and 2

Before proving Theorems 1 and 2, we first establish a lemma relating $\alpha_i^{(j)}$, $\beta_i^{(j)}$, and $\gamma_i^{(j)}$, followed by a proposition providing their closed-form expressions.

Lemma 2. *Let $\alpha_i^{(j)}$, $\beta_i^{(j)}$ and $\gamma_i^{(j)}$ be as in* (3.11). *For $j=2,\ldots,k$ and $i=2,\ldots,d-1$, we have*

$$
\alpha_i^{(j)}+\beta_i^{(j)}=\gamma_i^{(j)}.
\tag{4.1}
$$

Proof. For $j=2$, we have $\alpha_i^{(2)}+\beta_i^{(2)}=d+1-i=\gamma_i^{(2)}$ for $i=2,3,\ldots,d$, which proves (4.1). For $j>2$, (4.1) follows inductively using that (4.1) holds for smaller values of j. □

Proposition 5 (Closed forms for $\alpha_i^{(j)}$, $\beta_i^{(j)}$, and $\gamma_i^{(j)}$)**.** *Let $d\in\mathbb{N}$ and let $j\geq 2$. Let $\alpha_i^{(j)}$, $\beta_i^{(j)}$ and $\gamma_i^{(j)}$ be as in* (3.11). *Then the following closed forms hold:*

$$
\begin{aligned}
\alpha_i^{(j)} &= \binom{d-i+\min\{j-2,i-1\}}{\min\{j-2,i-1\}}, && i=2,\ldots,d,\\
\beta_i^{(j)} &= \binom{d-1-i+\min\{j-1,i\}}{\min\{j-1,i\}}, && i=2,\ldots,d-1,\\
\gamma_i^{(j)} &= \binom{d-i+\min\{j-1,i\}}{\min\{j-1,i\}}, && i=2,\ldots,d.
\end{aligned}
$$

Proof. We prove the formulas for $\gamma_i^{(j)}$ and $\alpha_i^{(j)}$ by induction on j; the statement for $\beta_i^{(j)}$ follows by the same argument with d replaced by $d-1$ (and with the upper summation index $d-1$).

Step 1: Closed form for $\gamma_i^{(j)}$. Define $r(j,i) := \min\{j-1, i\}$. We claim that

$$\gamma_i^{(j)} = \binom{d-i+r(j,i)}{r(j,i)} \qquad (i = 2,\dots,d,\ j \geq 2).$$

For $j = 2$ we have $r(2,i) = 1$ and

$$\binom{d-i+1}{1} = d+1-i = \gamma_i^{(2)}.$$

Now assume the claim holds for $j-1 \geq 2$ and fix $i \in \{2,\dots,d\}$.

If $i \leq j-2$, then $\gamma_i^{(j)} = \gamma_i^{(j-1)}$ by (3.11). Since $i \leq j-2$, one has $r(j,i) = i = r(j-1,i)$, hence

$$\gamma_i^{(j)} = \gamma_i^{(j-1)} = \binom{d-i+r(j-1,i)}{r(j-1,i)} = \binom{d-i+r(j,i)}{r(j,i)}.$$

If $i \geq j-1$, then by (3.11) and the induction hypothesis, for every $\ell \geq i$ we have $\ell \geq j-1$ and thus $r(j-1,\ell) = j-2$. Hence,

$$\gamma_i^{(j)} = \sum_{\ell=i}^{d} \gamma_\ell^{(j-1)} = \sum_{\ell=i}^{d} \binom{d-\ell+j-2}{j-2} = \sum_{t=0}^{d-i} \binom{t+j-2}{j-2},$$

where we used the substitution $t = d-\ell$ in the last equality. By [14, 1.5.2 Chu's Theorem],

$$\sum_{t=0}^{N} \binom{t+r}{r} = \binom{N+r+1}{r+1} \tag{4.2}$$

and hence (with $N = d-i$ and $r = j-2$),

$$\gamma_i^{(j)} = \binom{(d-i)+(j-2)+1}{(j-2)+1} = \binom{d-i+j-1}{j-1}$$

Since $i \geq j-1$ implies $r(j,i) = j-1$, this equals $\binom{d-i+r(j,i)}{r(j,i)}$ and the induction is complete.

Step 2: Closed form for $\alpha_i^{(j)}$. Define $s(j,i) := \min\{j-2, i-1\}$. We claim that

$$\alpha_i^{(j)} = \binom{d-i+s(j,i)}{s(j,i)} \qquad (i = 2,\dots,d,\ j \geq 2).$$

For $j = 2$ we have $s(2,i) = 0$ and $\binom{d-i}{0} = 1 = \alpha_i^{(2)}$. For the inductive step, the argument is identical to Step 1: in the range $i \leq j-2$ the values do not change, while in the range $i \geq j-1$ one obtains

$$\alpha_i^{(j)} = \sum_{\ell=i}^{d} \alpha_\ell^{(j-1)} = \sum_{t=0}^{d-i} \binom{t+j-3}{j-3} = \binom{d-i+j-2}{j-2},$$

again by (4.2) (with $N = d-i$ and $r = j-3$). This yields the claimed formula for α.

Step 3: Closed form for $\beta_i^{(j)}$. The recursion for β is the same as for γ with d replaced by $d-1$ and with indices $i \leq d-1$. Applying Step 1 with $d \mapsto d-1$ gives

$$\beta_i^{(j)} = \binom{(d-1)-i+\min\{j-1,i\}}{\min\{j-1,i\}}, \qquad i = 2,\dots,d-1.$$

This completes the proof. $\square$

Now we can prove Theorem 1.

Proof of Theorem 1. By the results of the previous section, to prove Theorem 1, it suffices to find the optimal value of (3.19). First we prove the following claim.

Claim 1. Their exists an optimal solution $(y_1^*, y_2^*, y_3^*, y_4^*)$ to the linear program (3.19) which satisfies $y_2^* = y_3^*$.

Proof of Claim 1. Let $(y_1^*, y_2^*, y_3^*, y_4^*)$ be an optimal solution to (3.19). If $y_2^* > y_3^*$, then replacing y_3^* with y_2^* it is clear that $(y_1^*, y_2^*, y_2^*, y_4^*)$ satisfies all constraints and the value of the objective function remains unchanged. ■

Using Claim 1 and Lemma 2, the linear program (3.19) simplifies to:

$$
\begin{aligned}
\max_{y_1,y_4} \quad & -y_1-(d-1)y_4,\\
\text{subject to} \quad & y_1-y_2+dy_4\geq 0,\\
& y_2-y_4\geq 0,\\
& \gamma_i^{(k)}(y_2-y_4)\geq (-1)^{d+1-i}\binom{d-k}{i-k} \quad \text{for } i=k,k+1,\dots,d,\\
& y_1\geq 0,\ y_2\geq 0,\ y_4\geq 0.
\end{aligned}
\tag{4.3}
$$

Claim 2. There is an optimal solution (y_1^*,y_2^*,y_4^*) to the linear program (4.3) such that

$$y_2^*=y_1^*+dy_4^*.$$

Proof of Claim 2. Let (y_1^*,y_2^*,y_4^*) be an optimal solution to (4.3). If $y_2^*<y_1^*+dy_4^*$, then we can replace y_2^* with $y_1^*+dy_4^*$, still satisfying all constraints of (4.3) and not changing its objective function. This proves Claim 2. ■

Using Claim 2, (4.3) simplifies to

$$
\begin{aligned}
\max_{y_1,y_4} \quad & -y_1-(d-1)y_4,\\
\text{subject to} \quad & \gamma_i^{(k)}(y_1+(d-1)y_4)\geq (-1)^{d+1-i}\binom{d-k}{i-k} \quad \text{for } i=k,k+1,\dots,d,\\
& y_1\geq 0,\ y_4\geq 0.
\end{aligned}
\tag{4.4}
$$

Clearly, the optimal solution satisfies

$$y_1^*+(d-1)y_4^*=\max\big(0,\max_{i\in k,\dots,d}(-1)^{d+1-i}\binom{d-k}{i-k}(\gamma_i^{(k)})^{-1}\big). \tag{4.5}$$

By Proposition 5, for every $i\in\{k,\dots,d\}$ we have

$$\gamma_i^{(k)}=\binom{d-i+k-1}{k-1}.$$

Substituting this into (4.5) yields

$$(-1)^{d+1-i}\binom{d-k}{i-k}\big(\gamma_i^{(k)}\big)^{-1}=(-1)^{d+1-i}\,\frac{\binom{d-k}{i-k}}{\binom{d-i+k-1}{k-1}}.$$

A straightforward factorial simplification gives

$$\frac{\binom{d-k}{i-k}}{\binom{d-i+k-1}{k-1}}=\frac{\binom{d-1}{i-k}}{\binom{d-1}{k-1}}, \qquad i=k,\dots,d. \tag{4.6}$$

Indeed,

$$\frac{\binom{d-k}{i-k}}{\binom{d-i+k-1}{k-1}}=\frac{(d-k)!(k-1)!}{(i-k)!(d-i+k-1)!}=\frac{\binom{d-1}{i-k}}{\binom{d-1}{k-1}},$$

where in the last equality we used $d-1-(i-k)=d-i+k-1$. Consequently, (4.5) becomes

$$y_1^*+(d-1)y_4^*=\max\left(0,\ \frac{1}{\binom{d-1}{k-1}}\max_{i=k,\dots,d}(-1)^{d+1-i}\binom{d-1}{i-k}\right). \tag{4.7}$$

Writing $r:=i-k\in\{0,\dots,d-k\}$, we may equivalently rewrite the inner maximum as

$$\max_{r=0,\dots,d-k}(-1)^{d+1-k-r}\binom{d-1}{r}. \tag{4.8}$$

Since $\binom{d-1}{r} > 0$, the sign is determined solely by the parity of $d+1-k-r$; hence only indices r with $d+1-k-r$ even can contribute to the maximum. Let r_0 be a maximizer in this parity class and set $i_0 := k + r_0$. Then

$$\max_{i=k,\dots,d} (-1)^{d+1-i}\binom{d-1}{i-k} = \binom{d-1}{r_0},$$

and therefore

$$y_1^* + (d-1)y_4^* = \frac{\binom{d-1}{r_0}}{\binom{d-1}{k-1}}.$$

Next we derive the formula for the optimizer r_0. The sequence $b_r := \binom{d-1}{r}$ is strictly increasing for $r \le \lfloor (d-1)/2 \rfloor$ and strictly decreasing afterwards (with a two-point plateau at the top if d is even). In (4.8), we maximize b_r over the truncated parity class

$$\{\, r : 0 \le r \le d-k,\ r \equiv d+1-k \pmod 2 \,\}.$$

Case 1: $d-k \le \lfloor \frac{d-1}{2} \rfloor$. Since (b_r) is increasing on $0 \le r \le \lfloor \frac{d-1}{2} \rfloor$, the maximum over $0 \le r \le d-k$ is attained at the largest admissible $r \le d-k$ of the required parity, which is $r_0 = d-k-1$.

Case 2: $d-k > \lfloor \frac{d-1}{2} \rfloor$. If d is even, the two equal maxima of the sequence b_r occur at $\frac{d}{2}-1$ and $\frac{d}{2}$, and exactly one has the required parity. Thus

$$r_0 = \begin{cases} \frac{d}{2}-1, & \text{if } \frac{d}{2}-k \text{ is even,} \\ \frac{d}{2}, & \text{if } \frac{d}{2}-k \text{ is odd.} \end{cases}$$

If d is odd, the unique maximum of the sequence b_r occurs at $\frac{d-1}{2}$; if $\frac{d-1}{2}$ has the wrong parity, the nearest admissible indices are $\frac{d-1}{2} \pm 1$, which yield the same value by symmetry. Thus

$$r_0 = \begin{cases} \frac{d-1}{2}, & \text{if } \frac{d+3}{2}-k \text{ is even,} \\ \frac{d-1}{2} \pm 1, & \text{if } \frac{d+3}{2}-k \text{ is odd.} \end{cases}$$

Summarizing the above cases, we see that r_0 is always the admissible index $r \equiv d-k-1 \pmod 2$ that is closest to the peak $r = \lfloor \frac{d-1}{2} \rfloor$ of the unimodal sequence $\binom{d-1}{r}$, unless the truncation $r \le d-k$ is active, in which case $r_0 = d-k-1$. This can be written equivalently in the compact form

$$r_0 = \begin{cases} d-k-1, & d-k \le \lfloor \frac{d-1}{2} \rfloor, \\ \lfloor \frac{d-1}{2} \rfloor + (-1)^d \, \dfrac{1-(-1)^{\Delta}}{2}, & d-k > \lfloor \frac{d-1}{2} \rfloor, \end{cases} \qquad \Delta := d-k-1-\left\lfloor \frac{d-1}{2} \right\rfloor.$$

It remains to prove the realization of an optimal solution. Let $i_0 := k + r_0$. By complementary slackness applied to the optimal primal–dual pair of (3.12) and (3.19), we have

$$(\delta_j^{(i)})^* = 0 \quad \text{for all } i \ne k_0, \qquad (\delta_j^{(k)})^* = 0 \quad \text{for all } j \ne i_0, \qquad (\delta_{i_0}^{(k)})^* = (\gamma_{i_0}^{(k)})^{-1}. \tag{4.9}$$

Since there exists an optimal dual solution with $y_i^* \ne 0$ for $i = 1,2,3,4$ (any pair $(y_1, y_4) \in \mathbb{R}^2_{\ge 0}$ satisfying $y_1 + (d-1)y_4 = w_{k,d,-}$ is optimal), complementary slackness implies that the corresponding primal solution satisfies four equalities. Consequently,

$$\mathcal{B} = [a,1]^d, \qquad a = \frac{\alpha_{i_0}^{(k)}}{\gamma_{i_0}^{(k)}}, \qquad Q(z) = q_{d-S(z)}, \quad z \in \mathrm{Vert}(\mathcal{B}), \tag{4.10}$$

where the coefficients $(q_i)_{i=0}^d$ satisfy $\delta_j^{(i)} = \sum_{\ell=0}^{i} \binom{i}{\ell}(-1)^{\ell} q_{j-\ell}$. Using Proposition 5 and the fact that $i_0 \ge k$, we obtain

$$\gamma_{i_0}^{(k)} = \binom{d-i_0+k-1}{k-1}, \qquad \alpha_{i_0}^{(k)} = \binom{d-i_0+k-2}{k-2},$$

and hence

$$a = \frac{\alpha_{i_0}^{(k)}}{\gamma_{i_0}^{(k)}} = \frac{k-1}{d-1-r_0}.$$

The conditions (4.9) uniquely determine $(q_i)_{i=0}^d$ as the solution of a finite-difference equation of order k with initial conditions $q_0 = \cdots = q_{i_0-1} = 0$ and $q_{i_0} = (\gamma_{i_0}^{(k)})^{-1}$. Solving this recurrence yields

$$q_i = \frac{\binom{i-r_0-1}{k-1}}{\binom{d-1-r_0}{k-1}}, \qquad i = 0, \ldots, d,$$

with the convention $\binom{n}{m} = 0$ for $n < m$. Indeed, by (3.10) we have $\delta_i^{(1)} = \nabla q_i$ and inductively $\delta_i^{(j)} = \delta_i^{(j-1)} - \delta_{i-1}^{(j-1)} = \nabla^j q_i$, so $\delta_i^{(k)} = \nabla^k q_i$ is the k-th backward difference of (q_i). Since $(\delta_j^{(k)})^* = 0$ for all $j \geq i_0+1$, it follows that $\nabla^k q_i = 0$ for all $i \geq i_0+1$. The solutions of $\nabla^k q_i = 0$ are precisely the sequences of the form $q_i = \sum_{m=0}^{k-1} c_m \binom{i}{m}$, because $\nabla\binom{i}{m} = \binom{i-1}{m-1}$ and hence $\nabla^k\binom{i}{m} = 0$ for $m \leq k-1$ (see, e.g., [15, Sec. 2.6]). Using the initial conditions $q_{i_0-1} = \cdots = q_{i_0-k} = 0$ and $q_{i_0} = (\gamma_{i_0}^{(k)})^{-1}$, the expansion reduces to a single term, yielding $q_i = C\binom{i-i_0+k-1}{k-1} = C\binom{i-r_0-1}{k-1}$ with $C = q_{i_0} = (\gamma_{i_0}^{(k)})^{-1}$. Finally, since $\gamma_{i_0}^{(k)} = \binom{d-1-r_0}{k-1}$, we get $C = 1/\binom{d-1-r_0}{k-1}$.

Substituting these expressions into (4.10) completes the construction of the realizing quasi-copula Q. □

Proof of Theorem 2. The proof is analogous to the proof of Theorem 1. □

5. Concluding remarks and future research

In this paper, we addressed the problem of determining the extreme values of the mass distribution associated with a multidimensional k-increasing d-variate quasi-copula for $k = 2, 3, \ldots, d$, building on the linear programming approach that solved the case $k = 1$ (see [8] for $d = 3$, [9] for $d = 4$, [10] for $d \leq 17$, and [11] for the general case). We derived closed formulas for the extreme volumes as well as an explicit realization of one corresponding quasi-copula. Besides the symmetrization trick from [11], the main novelty enabling us to solve the linear programs was the introduction of new sets of variables, which significantly reduced their complexity. We also provided a numerical analysis of the results for dimensions $d \leq 15$.

Finally, we outline several directions for future research. Very recently, Anzilli and Durante [3] derived bounds on the average F-volume of closed rectangles in $[0,1]^2$ for a supermodular aggregation function F. In line with these results, it would be interesting to study the behavior of extreme volumes for supermodular aggregation functions [16] in higher dimensions, as well as for ultramodular and modular quasi-copulas investigated in [12].

It is clear that every extreme point (in the Krein–Milman sense) of the convex set of k-increasing quasi-copulas must be a maximal-volume one. Our construction of the extremal maximal-volume solution is symmetric—due to the symmetrization step—and is therefore not extreme. To better understand the geometry of the convex set of k-increasing quasi-copulas, it would be natural to characterize their extreme points. For semilinear copulas, such a characterization was obtained in [17].

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Microsoft Copilot to assist in deriving closed-form expressions for quantities defined recursively in (3.11). This assistance contributed to Proposition 5 and, consequently, to the closed-form solutions in Theorems 1 and 2. After using this tool, the authors reviewed and edited the content as necessary and take full responsibility for the contents of the published article.

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